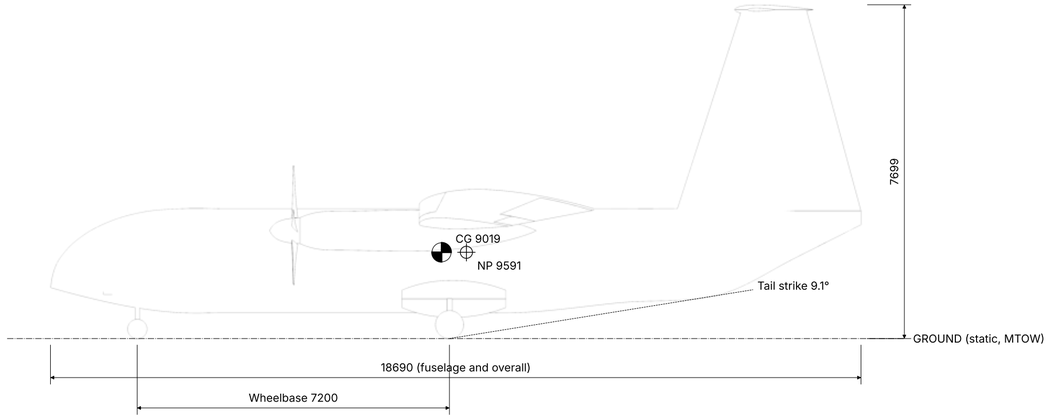
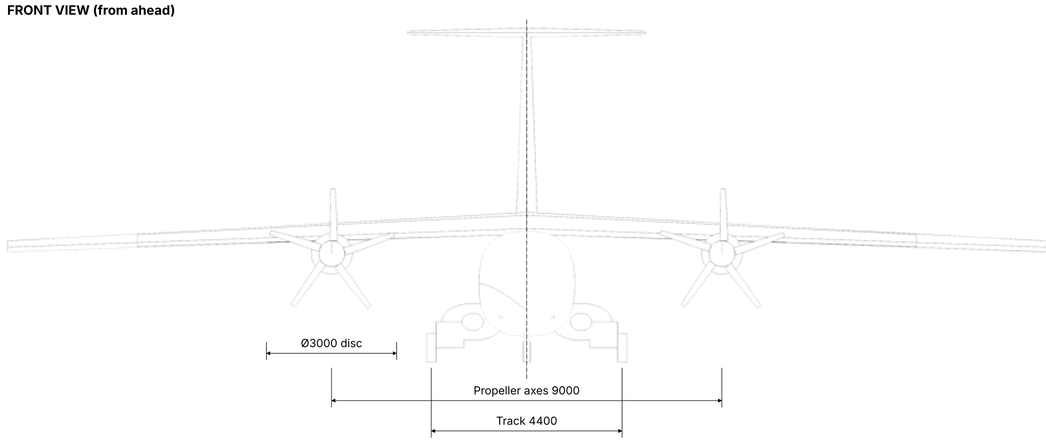
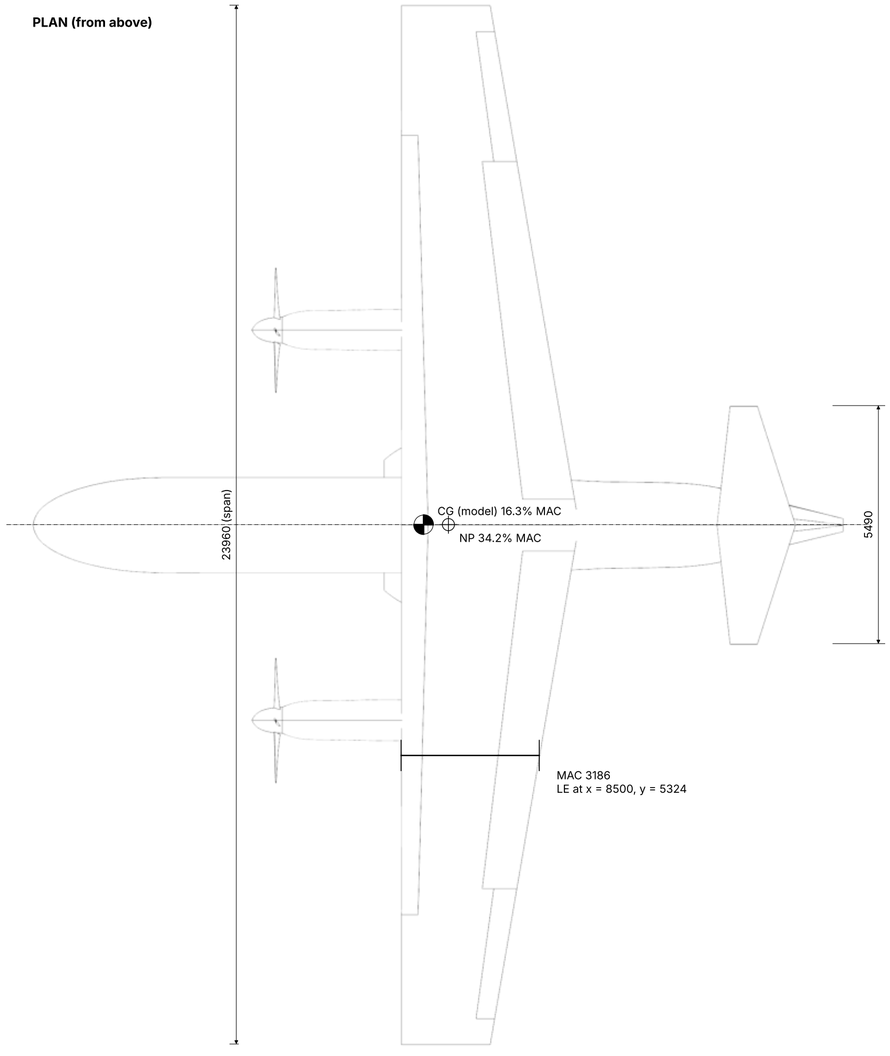
ELEVATION (from port)



FRONT VIEW (from ahead)



PLAN (from above)



DESIGN CHECKS (dimensions in mm; analysis/design_checks.py)

CHECK	MODEL	REFERENCE	RESULT
Span	23960	23960	+0.00 %
Wing area	73.60 m²	73.60 m²	+0.00 %
Aspect ratio	7.800	7.800	-0.00 %
MAC	3186	3186 (2/3 c_r (1+λ+λ²)/(1+λ))	+0.00 %
Prop tip to ground	991	>= 178 (14 CFR 23.925(a))	PASS
Prop tip to fuselage	1956	>= 25 (14 CFR 23.925(d))	PASS
Tail-strike angle	9.1°	report gives none	-
Tip-back, CG (model)	5.2°	>= 15° (Raymer)	FAIL
Tip-back, stated CG 9320	-3.5°	>= 15° (Raymer)	FAIL
CG (model), MTOW	9019, 16.3 % MAC	9320, 26.99 % (report)	-301 mm
Fuel-burn CG shift	195 fwd	≈180 fwd (report)	-
Static margin, model	18.0 % MAC	10.15 % (report)	-

TITLE	IATS hydrogen-electric STOL aircraft: general arrangement
DRAWING No.	AC-01-GA-001
REVISION	A (first issue)
SCALE	1:100 on A2
PROJECTION	First angle
UNITS	Dimensions in mm. Do not scale.
DOCUMENT TYPE	General arrangement (reference only, not for manufacture)
CREATED BY	G. Olu-Osagie
DATE OF ISSUE	2026-10-24
APPROVED	-
SHEET	1 of 1
MODEL	Autodesk Fusion: Engineering Portfolio / AC-01 IATS Aircraft
PARAMETERS	params/IATS_master_parameters.csv (value, unit, source file and line, status)

VALUES AND SOURCES (team files in University/3rd Year/Semester 2/Aircraft Design/)

Span 23960, area 73.60 m², AR 7.80, taper 0.50: wing summary L20 to L23 (C-1, C-2)
Root NACA 4412, tip NACA 23012, -2° dihedral, 0° LE sweep: wing summary L27 to L30
Wing root LE x = 8500: initial_sizing.m L51. Incidence 1.34°: wing summary L45 (A-1)
Tail per Table 5 (tail_design_summary.txt L4 to L16), sweep per plot_tail_geometry.m
Fuselage 18690: report p5 (C-3). Cross-section scaled from report Fig 2 (A-3)
Propeller Ø3000, 5 blades, twist per report p15. Nacelles at y = ±4500 (mass statement 4.1)
MAC 3186 by (2/3)c_r(1+λ+λ²)/(1+λ). Team scripts used b/AR = 3038 (C-6)
CG (model) x = 9019 from Table 3 masses at CAD centroids; stated 9320 (C-7)
NP from h_n = 0.25 + (a_t/a_w) V_H (1 - dε/dα), V_H = 0.198 with the measured tail arm 6938
Main gear x = 9200, half-track 2200: mass statement 5.1. Keel 600 above ground (A-7)
Every value: params/IATS_master_parameters.csv. Conflicts: docs/conflicts.md

Sizing and stability were my workstream on a six-person team project.
This model is my own interpretation of the team's design.